

HybridVAE

Semi-Supervised Network Intrusion Detection for Resource-Constrained Edge Devices

Scalable Computing for Data Analytics (CS6423)

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Group 2

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1. Abstract

This report presents HybridVAE, a semi-supervised neural network architecture designed for network intrusion detection on resource-constrained edge devices. The model integrates a Variational Autoencoder (VAE) path for anomaly detection with a parallel classification encoder, enabling simultaneous identification of known attack types and detection of previously unseen (zero-day) threats within a single forward pass. The system is trained and evaluated on the CICIDS2017 benchmark dataset comprising over 2.5 million network flow records across nine consolidated attack categories ^[1].

The HybridVAE achieves a Macro F1-score of 0.8702 and an anomaly detection AUROC of 0.8740 on the nine-class classification task, combining both capabilities in a model of only 2.92 MB. In a simulated zero-day scenario where the Heartbleed attack class is withheld from training, the HybridVAE successfully flags all unseen attack samples as anomalous, while a baseline XGBoost classifier confidently misclassifies them as benign traffic with 99.98% confidence.

To enable deployment on edge hardware, two compression strategies are applied: knowledge distillation ^[4] reduces the model to 221,462 parameters (0.894 MB) while retaining 99.1% of the teacher's Macro F1, and Iterative Magnitude Pruning (IMP) at 60% sparsity achieves a Macro F1 of 0.8992, surpassing the unpruned baseline. These results demonstrate that dual-path architectures can deliver both classification accuracy and anomaly detection capability at model sizes suitable for IoT gateways, embedded controllers, and other edge platforms where cloud connectivity is unavailable. ^[5]

2. Introduction

2.1 Problem Context

The proliferation of Internet of Things (IoT) devices across industrial, commercial, and domestic environments has created a vast and growing attack surface for network intrusions. Traditional intrusion detection systems (IDS) rely on either signature-based matching, which fails against novel attacks, or cloud-based machine learning inference, which requires persistent network connectivity and introduces latency. Edge devices such as IoT sensors, home gateways, and embedded controllers frequently operate on air-gapped or intermittently connected networks, making cloud-dependent detection architectures impractical.

These devices also face severe computational constraints. Limited RAM, storage, and CPU cycles preclude the deployment of large-scale models or ensemble methods that dominate tabular classification benchmarks. A deployable edge IDS must therefore be compact, capable of real-time inference without a GPU, and able to operate entirely offline.

2.2 The Zero-Day Challenge

A purely supervised classifier, no matter how accurate on known attack types, is fundamentally blind to novel threats. New attack vectors emerge continuously, and a model trained exclusively to classify known categories will assign confident but incorrect labels to unseen attacks. This represents a critical vulnerability in security systems: a classifier that confidently labels a Heartbleed exploit as benign traffic provides a false sense of security that is arguably worse than having no detector at all.

Anomaly detection methods based on autoencoders offer an alternative by learning the manifold of normal traffic and flagging deviations. However, standalone autoencoders cannot name the specific attack class, limiting their operational utility. The security community therefore requires models that provide both supervised classification of known threats and unsupervised anomaly detection for unknown threats.^[8]

2.3 Objectives

This project addresses the above challenges with the following objectives: design a hybrid architecture that unifies supervised classification and unsupervised anomaly detection in a single model; demonstrate zero-day detection capability through a withheld-class experiment; compress the model for edge deployment using knowledge distillation and iterative magnitude pruning; and benchmark the hybrid approach against pure supervised (XGBoost) and pure anomaly detection (Vanilla Autoencoder) baselines.

3. Dataset

3.1 CICIDS2017 Overview

The Canadian Institute for Cybersecurity Intrusion Detection System 2017 (CICIDS2017) dataset is the most widely cited modern IDS benchmark. It captures five days of realistic network traffic amounting to 2.83 million raw records, containing both benign activity and eleven labelled attack types. The dataset provides 77 network flow features extracted using CICFlowMeter, including packet-level statistics, flow duration metrics, flag counts, and inter-arrival time distributions.^[1]

3.2 Class Distribution

After cleaning (removal of infinite values, NaN imputation, and outlier clipping), the dataset contains 2,522,362 records. The class distribution exhibits extreme imbalance, with BENIGN traffic constituting 83.12% of all records while rare classes such as Infiltration (36 samples) and Heartbleed (11 samples) represent less than 0.01%.

Table 1: CICIDS2017 Class Distribution After Cleaning

Class	Count	Percentage (%)
BENIGN	2,096,484	83.12
DoS	193,748	7.68
DDoS	128,016	5.08
PortScan	90,819	3.60
BruteForce	9,152	0.36
Web Attack	2,143	0.09
Botnet	1,953	0.08
Infiltration	36	0.00
Heartbleed	11	0.00

3.3 Class Consolidation

The original eleven classes are consolidated to nine by merging the three Web Attack sub-types (Brute Force, XSS, and SQL Injection) into a single Web Attack category. This consolidation is motivated by the extremely low combined sample count (fewer than 700 samples across three sub-types), which produces unstable per-class F1 scores when kept separate. The merging reduces classifier noise without sacrificing semantic information, as all three sub-types share the same operational response profile.

3.4 Data Splitting

The dataset is split into 70% training, 15% validation, and 15% test sets using stratified sampling to ensure all classes are represented proportionally in each split regardless of their frequency. This stratification is critical for rare classes such as Heartbleed, which has only 11 total samples.

4. Methodology

4.1 Feature Preprocessing

4.1.1 Cleaning and Normalisation

Raw features undergo the following preprocessing pipeline: non-numeric columns and metadata fields (Flow ID, IP addresses, ports, timestamps) are removed; infinite values are replaced with NaN and subsequently imputed with the column median; extreme outliers beyond the 99.9th percentile are clipped to reduce the influence of anomalous feature values on the learned representation.

4.1.2 ZCA Whitening Transform

A Zero-phase Component Analysis (ZCA) whitening transform is applied to decorrelate features and normalise their variance. Crucially, the whitening transform is fitted exclusively on BENIGN training samples. This implements the one-class principle underlying the anomaly detection capability: the resulting coordinate system encodes what normal traffic looks like, and attack traffic deviates measurably from this space. Mathematically, for a feature matrix X with covariance Σ :

$$X_{\text{whitened}} = X \cdot \Sigma^{-1/2}$$

Because the transform is fitted on BENIGN-only data, the VAE decoder's reconstruction error is calibrated to the normal traffic distribution. Any input that deviates from this learned manifold produces high reconstruction error, forming the basis of anomaly detection.

4.1.3 SMOTE Oversampling

Synthetic Minority Over-sampling Technique (SMOTE) is applied to the training set to address the extreme class imbalance. Minority attack classes are oversampled by generating synthetic samples through interpolation in feature space. This ensures the classifier receives sufficient gradient signal from rare classes during training, preventing the model from converging to a trivial BENIGN-majority solution.^[7]

4.2 HybridVAE Architecture

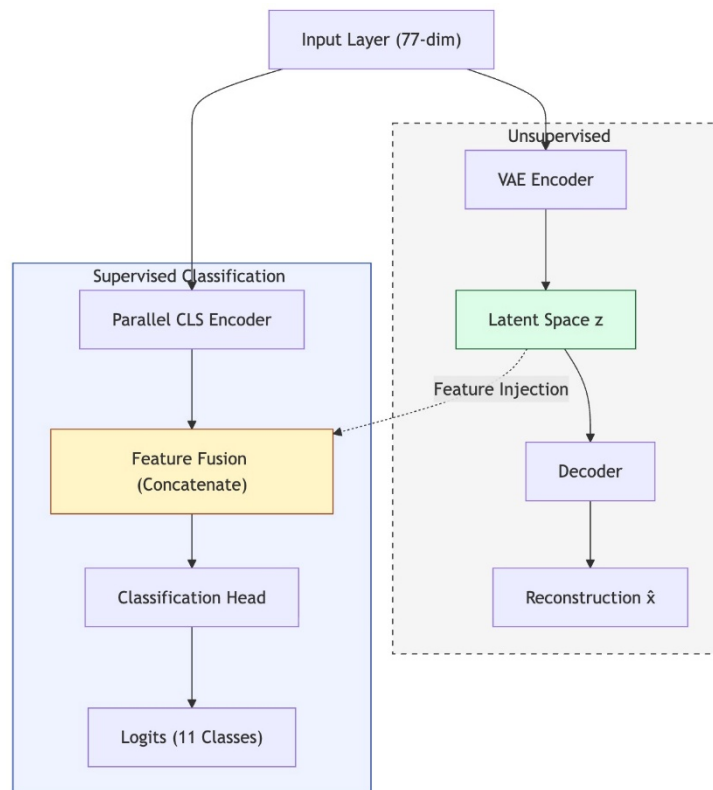


Figure 1: HybridVAE architecture diagram showing the dual-path encoder (VAE path and CLS path), latent space, decoder, and classifier head (from Slide 7)

The HybridVAE implements a dual-path encoder design that runs two parallel feature extraction streams from the same input:

VAE Path: The input passes through a three-layer encoder ($512 \rightarrow 256 \rightarrow 128$ neurons) with BatchNorm, GELU activation, and 30% dropout at each layer, followed by a residual block. The encoder output is projected to mean (μ) and log-variance ($\log \sigma^2$) vectors, from which a 64-dimensional latent vector z is sampled via the reparameterisation trick. The decoder mirrors the encoder ($128 \rightarrow 256 \rightarrow 512$) and reconstructs the input. This path learns the manifold of normal traffic; high reconstruction error at inference indicates anomalous input.^[2]

CLS Path: An independent discriminative encoder ($256 \rightarrow 256 \rightarrow 128$ neurons) processes the same input in parallel. This path is optimised purely for class separation and produces a 128-dimensional feature vector (`cls_feat`).

Classifier Head: The latent vector z (64-dim) and `cls_feat` (128-dim) are concatenated into a 192-dimensional representation, which is passed through a classification head to produce N-class logits. This fusion allows the classifier to leverage both the generative latent representation and the discriminative features for improved classification.

The complete model contains 749,270 trainable parameters and occupies 2.92 MB in serialised form.

4.3 Loss Function

The HybridVAE is trained with a multi-component objective function that serves three distinct purposes:

$$L_{total} = \alpha \cdot L_{recon} + \beta \cdot L_{cls} + \gamma \cdot L_{KL} \quad (\alpha = 0.20, \beta = 0.65, \gamma = 0.05)$$

Reconstruction Loss (L_{recon} , $\alpha = 0.20$): L1 (Mean Absolute Error) loss computed between the input and its reconstruction, trained exclusively on BENIGN samples. By restricting reconstruction to normal traffic, the decoder never learns to reconstruct attack patterns, preserving the anomaly detection capability even after joint training with the classifier.

Classification Loss (L_{cls} , $\beta = 0.65$): A weighted combination of Focal Loss (70%) and Label Smoothing Cross-Entropy (30%). Focal Loss down-weights easy, correctly-classified samples using a modulating factor $(1 - p_t)^\gamma$ with $\gamma = 1.5$, forcing the model to focus gradient updates on hard-to-classify minority classes. Class weights are set inversely proportional to class frequency. This is the dominant loss term, reflecting that attack classification is the primary task.^[3]

KL Divergence (L_{KL} , $\gamma = 0.05$): Regularises the latent distribution $q(z|x)$ to approximate a standard normal prior $N(0, I)$. The small weight (0.05) prevents posterior collapse while keeping the latent space smooth and continuous, which is essential for meaningful reconstruction error scoring.

4.4 Training Configuration

Table 2: HybridVAE Training Hyperparameters

Parameter	Value	Rationale
Optimiser	AdamW	Decoupled weight decay improves generalisation
Learning Rate	1e-3	Standard starting point; CosineAnnealing handles decay
Scheduler	CosineAnnealingLR	Smooth LR decay avoids destabilising VAE training
Batch Size	512	Stable gradient estimates on 1.7M samples
Epochs	75	One full cosine cycle; early stopping by best val Macro F1
Latent Dimension	64	Balanced bottleneck; 32 lost info, 128 under-compressed
Dropout Rate	0.3	Prevents overfitting on SMOTE-augmented data
Gradient Clipping	1.0	Prevents KL divergence explosions in early epochs
Focal γ	1.5	Moderate hard-example mining; $\gamma=2$ too aggressive

4.5 Evaluation Metrics

Accuracy alone is misleading when 83% of traffic is benign: a trivial model predicting BENIGN for every sample achieves 83% accuracy while catching zero attacks. The evaluation framework therefore prioritises the following metrics:

Macro F1-Score (primary): Averages F1 across all nine classes equally, so a rare class (Heartbleed with 11 samples) matters as much as BENIGN (2 million samples).

AUROC: Measures anomaly detection quality across all reconstruction error thresholds. Essential for evaluating the VAE path.

Weighted F1: Confirms volume-weighted performance has not degraded while optimising Macro F1.

Per-Class Precision and Recall: Precision catches over-alerting (false positives); recall catches missed attacks (false negatives).

5. Baseline Models

5.1 Vanilla Autoencoder

The Vanilla Autoencoder establishes a pure anomaly-detection baseline. It mirrors the HybridVAE's encoder-decoder architecture but without the variational bottleneck (no sampling, no KL divergence) and without the classification head. The model is trained exclusively on BENIGN traffic using L1 reconstruction loss.

The Vanilla AE contains 495,501 parameters and achieves a test AUROC of 0.7587 using the 95th percentile reconstruction error (0.492) as the anomaly threshold. At this threshold, 60.15% of attack samples (38,422 out of 63,882) are flagged as anomalous, meaning approximately 40% of attacks are missed. Critically, the Vanilla AE provides no classification capability: it can flag that something is abnormal but cannot identify what type of attack is occurring.

5.2 XGBoost Classifier

XGBoost represents the best achievable closed-set supervised performance on tabular data. It is configured with 500 estimators, max depth of 8, learning rate of 0.1, and trained with inverse-frequency class weights and early stopping (patience of 30 rounds). The model is trained on the same whitened features as the HybridVAE.^[6]

XGBoost achieves a Macro F1-score of 0.9162, the highest among all models evaluated. However, it provides zero anomaly detection capability: there is no mechanism to flag inputs that do not belong to any known class. The serialised model occupies 18.18 MB, making it approximately 6x larger than the HybridVAE.

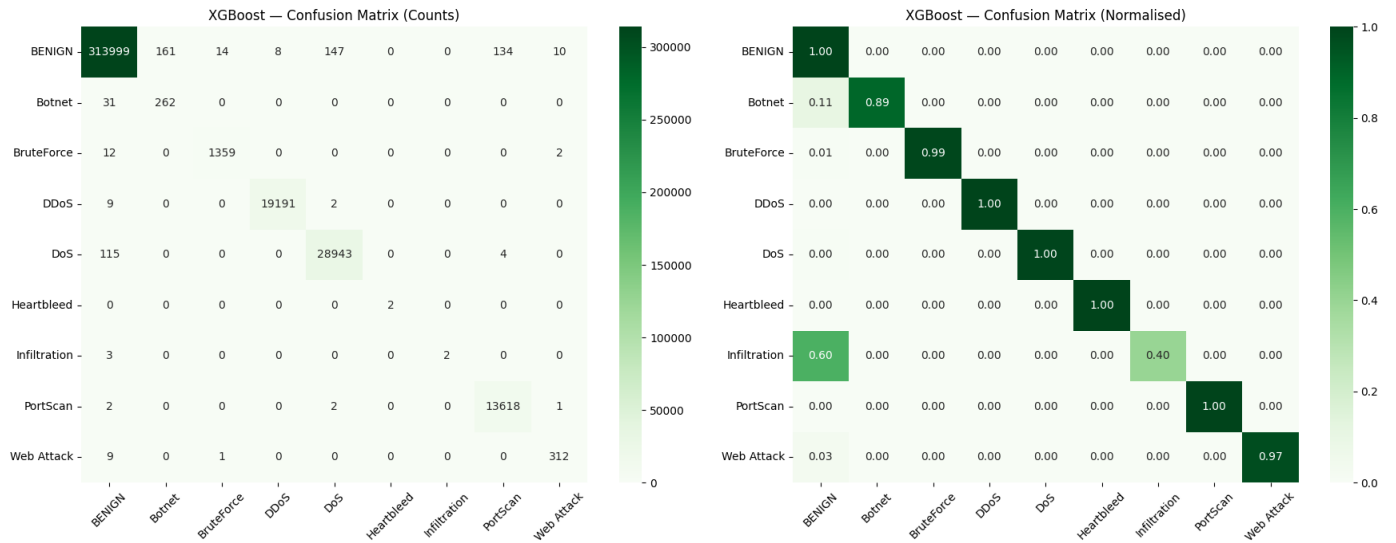


Figure 2: XGBoost confusion matrix (normalised) showing per-class classification performance (from Slide 12)

6. Experimental Results

6.1 HybridVAE Nine-Class Performance

The full HybridVAE trained on all nine classes achieves the following headline metrics: Macro F1-score of 0.8702, anomaly detection AUROC of 0.8740, and Weighted F1-score of 0.9899. The model occupies 2.92 MB with 749,270 parameters.

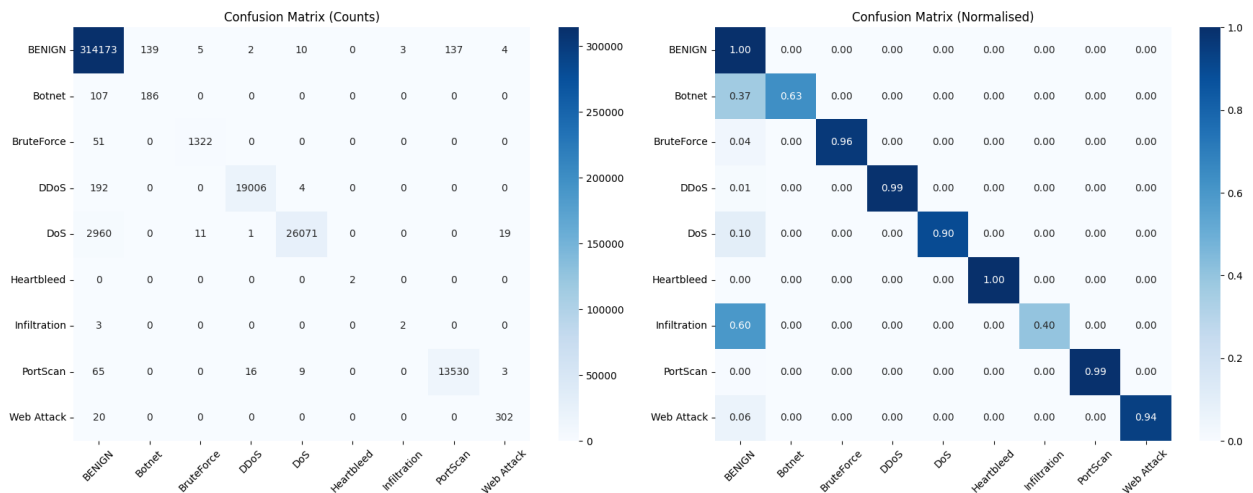


Figure 3: HybridVAE 9-class confusion matrix (counts and normalised) from Cell A6

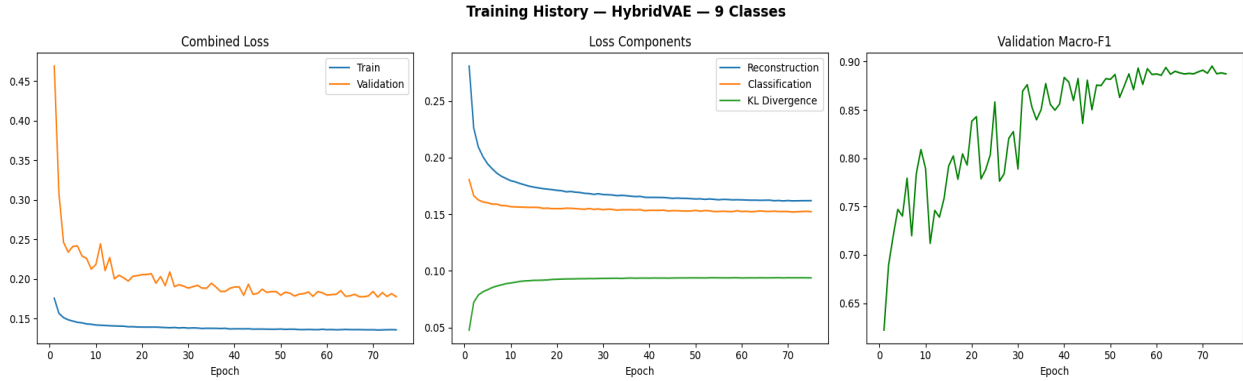


Figure 4: HybridVAE training curves — combined loss, loss components, and validation Macro F1 across 75 epochs (from Cell A5)

The Weighted F1 of 0.9899 confirms that volume-weighted performance remains excellent, while the Macro F1 of 0.8702 indicates strong per-class performance even for minority classes. The gap between Weighted and Macro F1 reflects the remaining difficulty in classifying extremely rare classes (Heartbleed, Infiltration) where very few training samples exist even after SMOTE augmentation.

6.2 Comparative Analysis

Table 3: Baseline Comparison

Model	Parameters	Size (MB)	Macro F1	AUROC	Anomaly?
Vanilla AE	495,501	1.93	—	0.7587	Yes (limited)
HybridVAE	749,270	2.92	0.8702	0.8740	Yes
XGBoost	N/A (trees)	18.18	0.9162	N/A	No

The comparative analysis reveals a clear trade-off landscape. XGBoost delivers the highest Macro F1 (0.9162) but is blind to novel attacks and occupies 6x more storage. The Vanilla AE provides basic anomaly detection (AUROC 0.76) but cannot classify attacks. The HybridVAE uniquely combines both capabilities: its classification performance approaches XGBoost (within 5 percentage points of Macro F1) while providing anomaly detection that substantially exceeds the Vanilla AE baseline (AUROC 0.874 vs 0.759).

7. Zero-Day Threat Detection

7.1 Experimental Design

To simulate a zero-day attack scenario, the Heartbleed class is entirely removed from the training and validation sets. Both the HybridVAE (model_d, 749,141 parameters) and XGBoost are retrained from scratch on the remaining eight classes using identical hyperparameters. At test time, the four Heartbleed samples in the test set are presented to both models as completely unseen attack traffic.

The anomaly detection threshold for the HybridVAE is calibrated at the 95th percentile of BENIGN validation reconstruction error, yielding a threshold of 0.648. Any test sample with reconstruction error exceeding this threshold is flagged as anomalous regardless of the classifier’s prediction.

7.2 Results

Table 4: Zero-Day Detection — XGBoost vs HybridVAE on Withheld Heartbleed Samples

Sample	XGBoost Prediction	Confidence	HybridVAE	True Label
0	BENIGN	99.98%	ANOMALY	Heartbleed
1	BENIGN	99.98%	ANOMALY	Heartbleed
2	BENIGN	99.98%	ANOMALY	Heartbleed
3	BENIGN	99.98%	ANOMALY	Heartbleed

XGBoost confidently misclassifies all four Heartbleed samples as BENIGN with 99.98% confidence. This represents the most dangerous failure mode in a security system: a silent failure where the system provides high-confidence incorrect assurance that no attack is occurring.

In contrast, the HybridVAE produces a mean reconstruction error of 3.74 on the withheld Heartbleed samples, compared to a BENIGN mean of 0.16. This exceeds the 95th percentile threshold (0.648) by a factor of 5.8x, resulting in all four samples being flagged as ANOMALY. The system alerts operators to the presence of an unknown threat even though it has never encountered Heartbleed during training.

Statistical caveat. The CICIDS2017 dataset contains only four Heartbleed samples in the validation and test sets combined, which is the entire pool of withheld examples available to this experiment. While the directional result is clean — every withheld sample is flagged by the HybridVAE and missed by XGBoost — a sample size of $n = 4$ has insufficient statistical power to support a quantitative detection-rate claim. This experiment should therefore be understood as a *qualitative proof of concept* demonstrating that the reconstruction-error pathway exposes a class never seen during training, rather than as a quantitative measurement of zero-day detection accuracy. A statistically robust evaluation would require either withholding multiple classes in a leave-one-out protocol across the full nine-class set, generating synthetic Heartbleed-like flow records, or sourcing additional Heartbleed traffic from complementary IDS datasets. These extensions are deferred to future work and discussed in Section 10.4.

7.3 Eight-Class Model Validation

The eight-class HybridVAE achieves a Macro F1 of 0.8538 and Weighted F1 of 0.9920 on the known classes, confirming that removing one class does not significantly degrade performance on the remaining eight. This validates that the withheld-class experiment is a fair test of zero-day detection capability rather than an artefact of a degraded model.

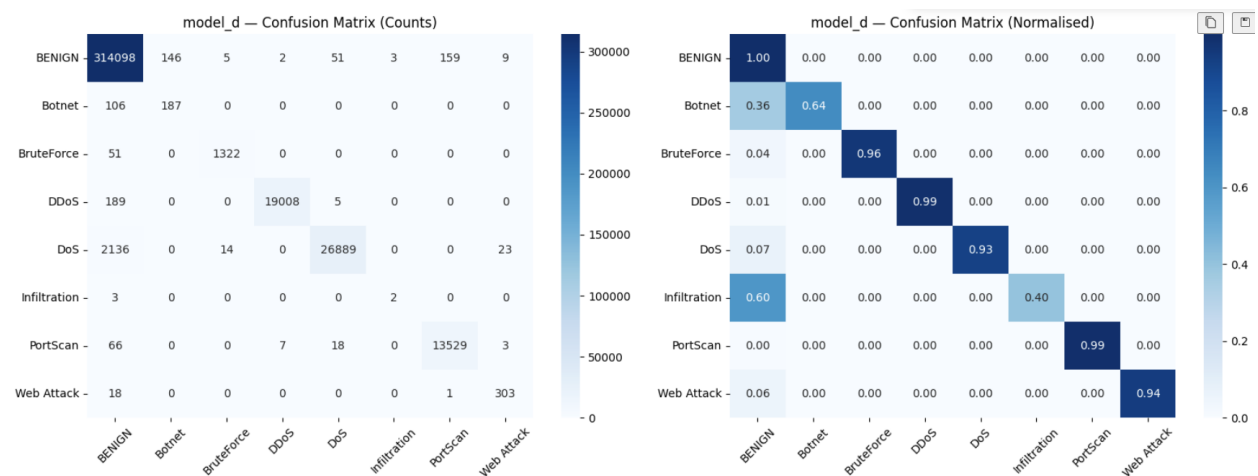


Figure 6: 8-class HybridVAE confusion matrix showing classification on the remaining eight classes (from Cell D5)

8. Model Compression for Edge Deployment

8.1 Knowledge Distillation

Knowledge distillation transfers the learned decision boundaries of a larger teacher model into a compact student network. The student HybridVAE has all hidden dimensions halved compared to the teacher: encoder dimensions of [256, 128, 64], latent dimension of 32, and CLS output dimension of 64, yielding 221,462 parameters (3.38x smaller than the teacher). The distillation uses temperature $T = 4.0$ and a distillation weight $\alpha = 0.7$ (70% soft labels from the teacher, 30% focal loss against hard labels).^[4]

8.1.1 Distillation Loss Function

Both distillation experiments (C-i and C-ii) use the same composite loss function. The student is trained with the standard HybridVAE objective, but the classification term is replaced with a weighted combination of distillation loss and supervised focal loss:

$$\begin{aligned} L_{total} &= 0.20 \cdot L_{recon} + 0.65 \cdot L_{cls} + 0.05 \cdot L_{KL} \\ L_{cls} &= 0.70 \cdot L_{distill} + 0.30 \cdot L_{focal_hard} \end{aligned}$$

The distillation term $L_{distill}$ is the Kullback–Leibler divergence between the teacher’s and student’s softmax outputs at temperature $T = 4$, scaled by T^2 to compensate for the gradient magnitude reduction introduced by softening:

$$L_{distill} = KL(\text{softmax}(\text{student} / T) \parallel \text{softmax}(\text{teacher} / T)) \times T^2$$

The teacher logits are detached from the computation graph so that gradients flow only through the student. Temperature $T = 4$ softens both probability distributions, exposing the inter-class similarity structure that hard labels discard. For example, a teacher prediction of 60% DoS / 35% DDoS / 5% other carries substantially more information than a one-hot “DoS” label, because it tells the student that DoS and DDoS are confusable classes that share underlying features. Learning to mimic this richer signal improves the student’s generalisation beyond what hard-label supervision alone provides.

Bridging the architectural mismatch. The distillation loss operates entirely in probability space rather than in weight or feature space. This is precisely how the framework accommodates radically different teacher architectures. Whether the teacher is a 749K-parameter HybridVAE (C-i) or an 18 MB ensemble of XGBoost decision trees (C-ii), the student only ever observes the teacher’s nine-class output probability vector. The student never accesses the teacher’s internal representations, intermediate activations, or learned weights. As long as both teacher and student produce probability distributions over the same nine classes, the distillation objective is well-defined and the architectural mismatch becomes irrelevant. The reconstruction loss L_{recon} (computed on BENIGN samples only) and the KL divergence L_{KL} are then computed independently of the teacher and preserve the student’s VAE anomaly detection capability. This separation explains the most striking result of experiment C-ii: the student inherits anomaly detection that the XGBoost teacher never possessed, because the VAE pathway is trained from data, not distilled from the teacher.

8.1.2 Experiment C-i: HybridVAE Teacher → Student

The student trained from the HybridVAE teacher achieves a Macro F1 of 0.8621 and anomaly AUROC of 0.8500, retaining 99.1% of the teacher’s Macro F1 in a model of only 0.894 MB. Because both teacher and student share the same architecture family, the student can directly learn the teacher’s latent-space structure through softened logits.

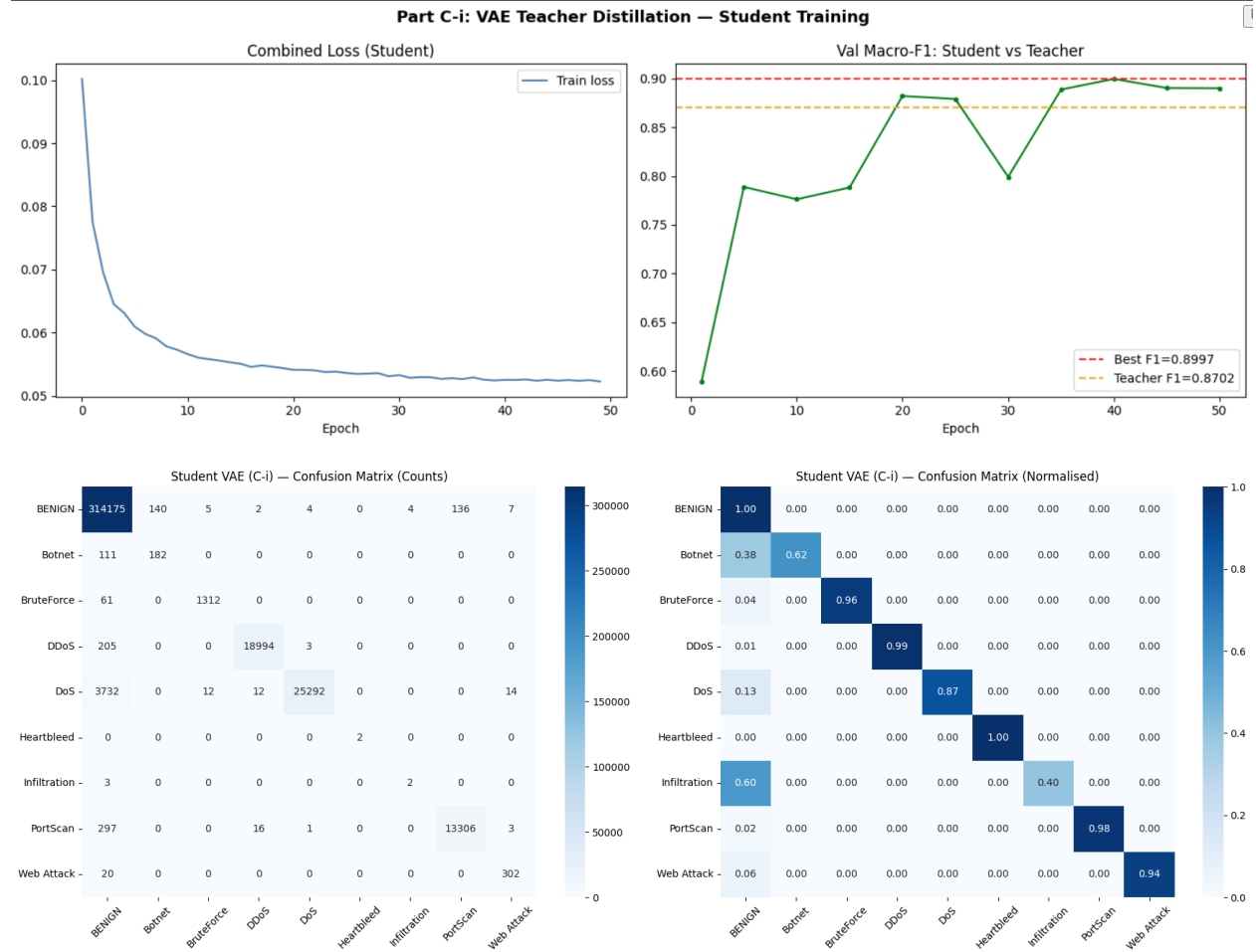


Figure 7: Student C-i confusion matrix showing distilled student per-class performance (from Cell C1-i)

8.1.3 Experiment C-ii: XGBoost Teacher → Student

Cross-architecture distillation transfers XGBoost’s gradient-boosted tree decision boundaries into the HybridVAE student. This achieves a Macro F1 of 0.8141, retaining 88.9% of XGBoost’s classification performance. Notably, the student gains anomaly detection capability that the XGBoost teacher never possessed, while compressing the model by 20x (from 18.18 MB to 0.894 MB).

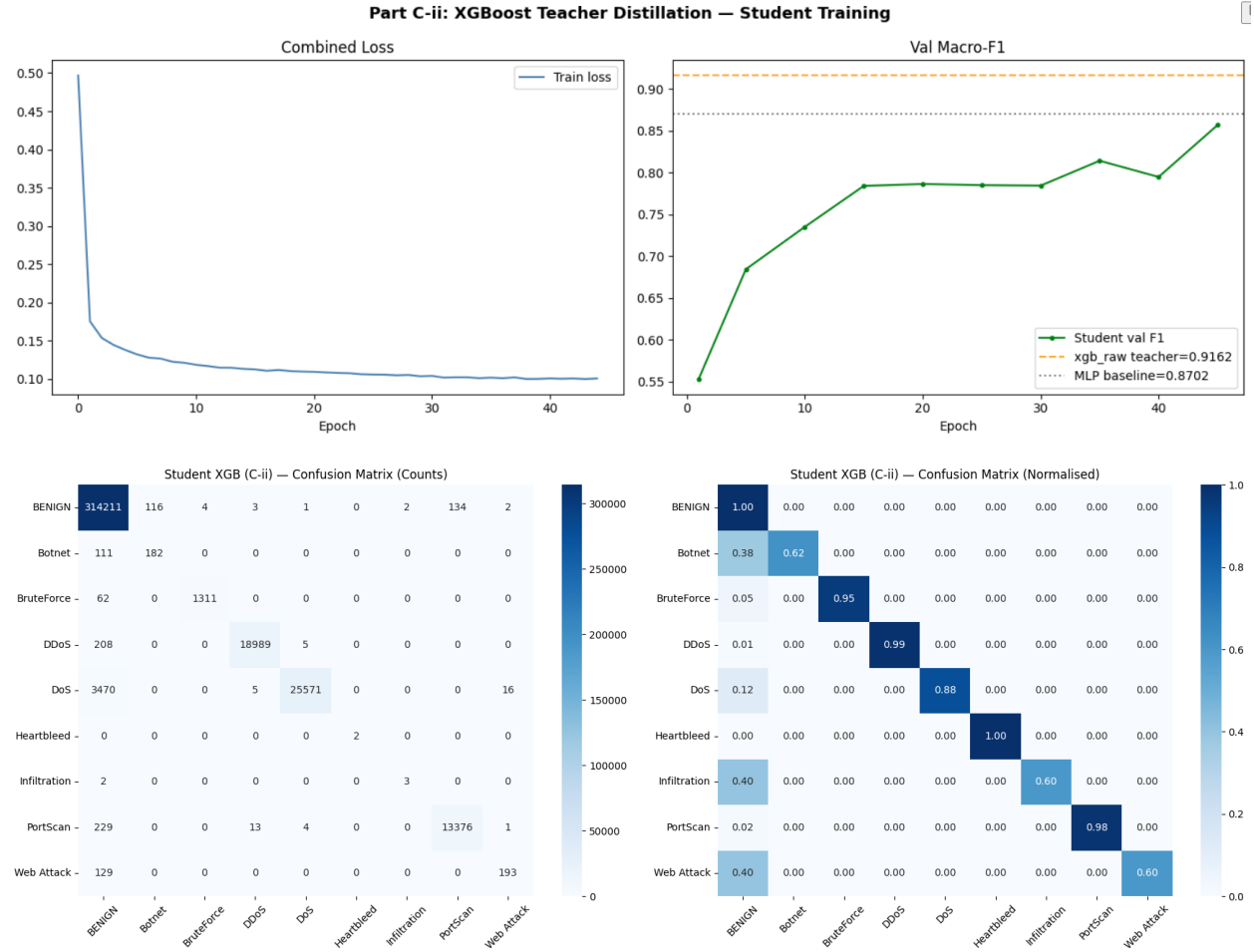


Figure 8: Student C-ii confusion matrix showing cross-architecture distillation performance (from Cell C1-ii)

8.1.4 Distillation Summary

Table 5: Knowledge Distillation Results

Model	Parameters	Size (MB)	Macro F1	AUROC	F1 Retained
HybridVAE Teacher	749,270	2.921	0.8702	0.8740	—
Student C-i (from VAE)	221,462	0.894	0.8621	0.8500	99.1%
XGBoost Teacher	N/A	18.176	0.9162	N/A	—
Student C-ii (from XGB)	221,462	0.894	0.8141	—	88.9%

8.2 Magnitude Pruning

8.2.1 One-Shot Pruning

One-shot magnitude pruning applies global L1 unstructured pruning at a specified sparsity level and evaluates immediately without retraining. This approach reveals how much redundancy exists in the trained model.

The implementation uses PyTorch’s built-in `torch.nn.utils.prune`^[9] library, applying `prune.global_unstructured()` with `prune.L1Unstructured` as the pruning method. Three properties of this configuration are critical to understanding the results. **Global** means the sparsity budget is allocated across all linear layers simultaneously, so layers containing predominantly small-magnitude weights lose more connections than layers with larger weights. **Unstructured** means individual weights are zeroed independently rather than entire neurons or filters being removed. **L1 magnitude** is the selection criterion: the smallest absolute-value weights are zeroed first. After pruning, `prune.remove()` bakes the zeros permanently into the weight tensors rather than retaining them as a soft mask, ensuring true model compression rather than masked simulation.

An important methodological note: pruning is applied to **every** `nn.Linear` layer in the model without exclusions, including the VAE encoder projections (`fc_mu`, `fc_log_var`), the decoder, the CLS encoder, and the classifier head. The decoder is therefore not explicitly protected from sparsification. Empirically, the AUROC drops only marginally (0.8740 at 0% sparsity to 0.8704 at 45% sparsity), suggesting the decoder retains sufficient capacity to model the BENIGN manifold even when its weights are subject to global pruning. Future work could investigate whether explicitly excluding the decoder from the pruning pool improves anomaly detection at higher sparsity levels.

Table 6: One-Shot Pruning Results

Sparsity	Macro F1	AUROC	Observation
0% (baseline)	0.8702	0.8740	Unpruned reference
20%	0.8703	0.8720	No degradation
40%	0.8729	0.8710	Slight improvement
45%	0.8773	0.8704	Best one-shot result
50%	0.7880	0.8680	Performance cliff
60%	0.6189	0.8636	Collapse

The results reveal a sharp performance cliff at 50% sparsity. Below this threshold, pruning has negligible impact (and slightly improves Macro F1 at 45%), while exceeding it causes rapid degradation. At 60%, the model collapses to 0.619.

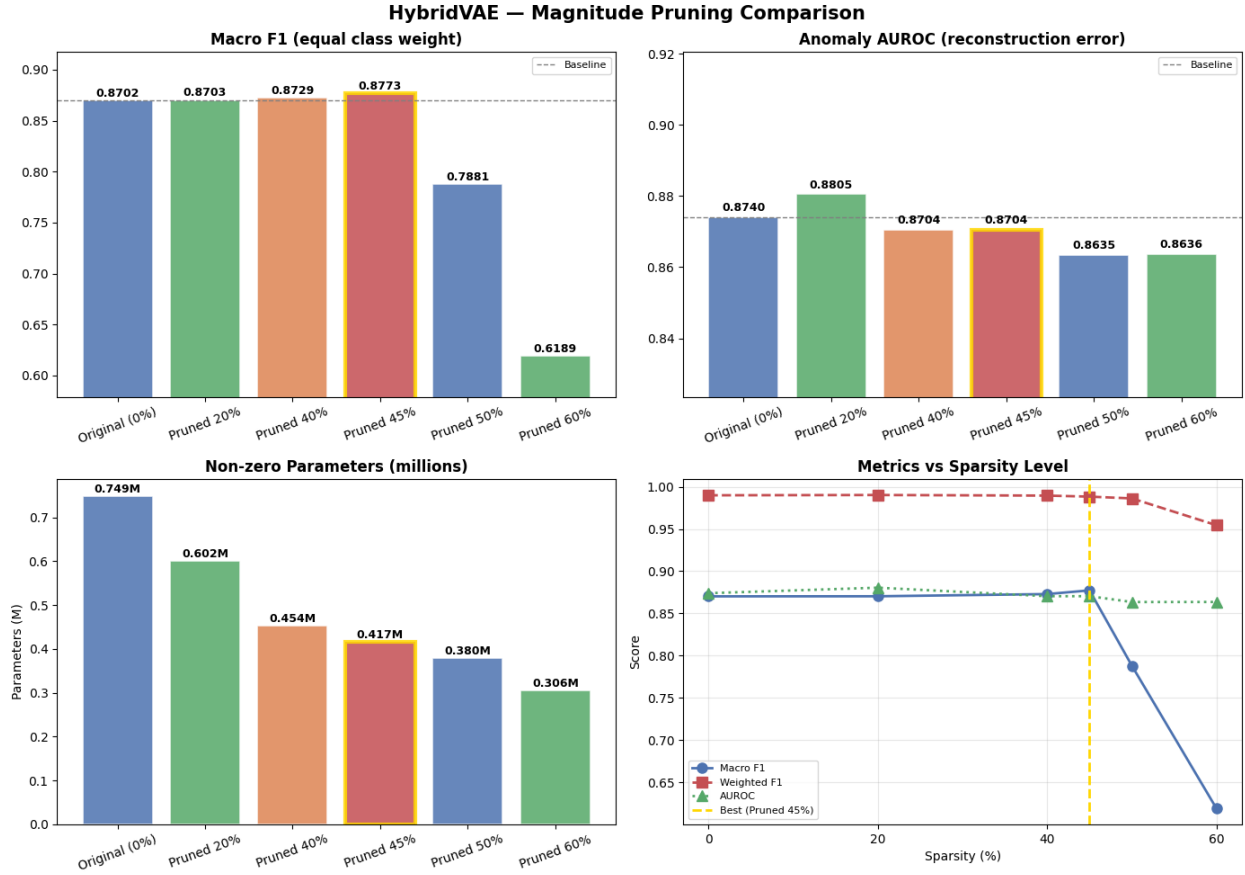


Figure 9: One-shot pruning curve — Macro F1 and AUROC vs sparsity level showing the cliff at 50% (from Cell F2)

8.2.2 Iterative Magnitude Pruning (IMP)

Inspired by the Lottery Ticket Hypothesis, Iterative Magnitude Pruning (IMP) finds sparse sub-networks within the dense model that can match or exceed the original’s accuracy. IMP operates by iteratively pruning small fractions and retraining between rounds (20 epochs per round, LR = $3e-4$). The sparsity schedule progresses through cumulative targets of 20%, 35%, 50%, 60%, and 70%.^[5]

Table 7: IMP vs One-Shot Comparison

Method	Sparsity	Macro F1	AUROC
Baseline (unpruned)	0%	0.8702	0.8740
One-shot (best)	45%	0.8773	0.8704
IMP (best)	60%	0.8992	0.8710
One-shot @ 60%	60%	0.6189	0.8636

IMP at 60% sparsity achieves a Macro F1 of 0.8992, surpassing the unpruned baseline (0.8702) by 2.9 percentage points. At the same 60% sparsity, one-shot pruning collapsed to 0.619 — a 28 percentage point gap. The iterative retraining allows the network to redistribute information across remaining weights, finding a more efficient parameterisation than the original dense model.

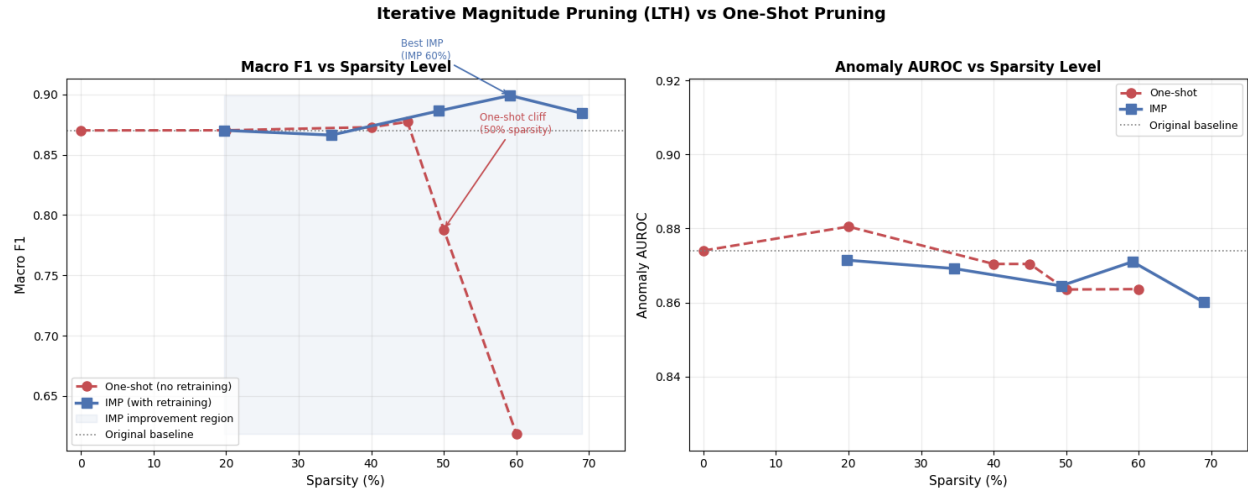


Figure 10: IMP vs one-shot comparison chart showing Macro F1 across sparsity levels (from Cell G2)

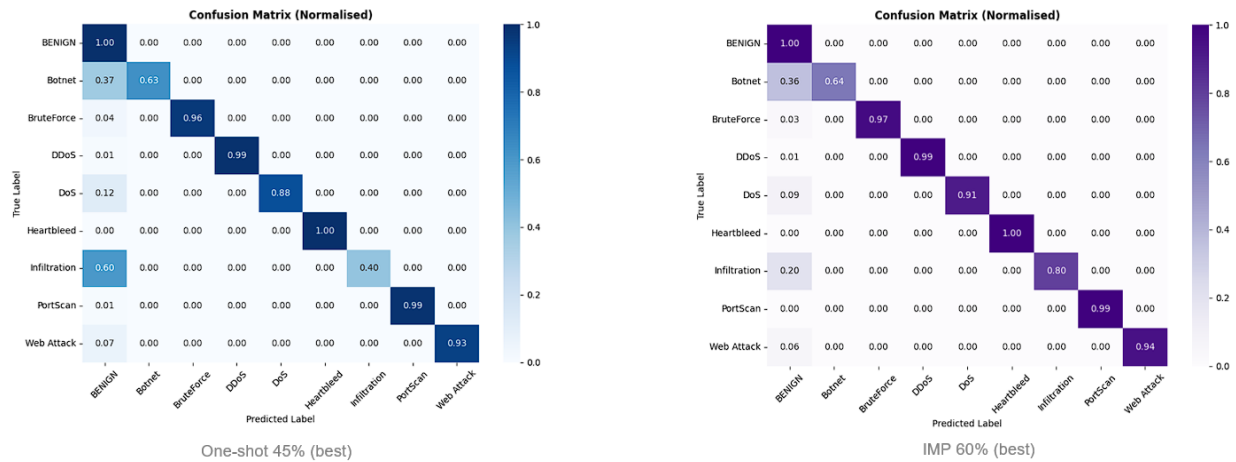


Figure 11: Confusion matrices: one-shot 45% vs IMP 60% side-by-side comparison (from Slides 26–27)

9. Team Contribution

Yashodhar Rathod: HybridVAE architecture design, knowledge distillation (experiments, 3.38× compression), Discussion & Conclusion sections, report and PPT coordination.

Vinyas Ravi: Loss function formulation (CE + KL + reconstruction), training pipeline (AdamW, CosineAnnealingLR, FocalLoss), evaluation harness, feature preprocessing, zero-day threat experiments, Introduction section.

Pouya Mohammadkhani: Magnitude pruning (one-shot & IMP), anomaly scoring pipeline, complete model comparison table, comparative analysis.

Vishvasundar: Vanilla Autoencoder baseline, dataset analysis, class consolidation, stratified data splitting, Appendix A.

Yingjun Sun: XGBoost baseline, hyperparameter tuning, per-class benchmarking, References section.

10. Complete Model Comparison

The following table summarises all models developed and evaluated, providing a comprehensive view of the accuracy-size-capability trade-off:

Table 8: Full Model Zoo

Model	Parameters	Size	Macro F1	AUROC	Anomaly?
Vanilla AE	495K	1.93 MB	—	0.759	Yes
HybridVAE (teacher)	749K	2.92 MB	0.870	0.874	Yes
XGBoost (teacher)	trees	18.18 MB	0.916	—	No
Student C-i (VAE→)	221K	0.89 MB	0.862	0.850	Yes
Student C-ii (XGB→)	221K	0.89 MB	0.814	—	Yes
One-shot 45%	~416K	~1.53 MB	0.877	0.870	Yes
IMP 60%	~306K	~1.21 MB	0.899	0.871	Yes

The best edge deployment candidate is the IMP 60% model: highest Macro F1 of any HybridVAE variant (0.899), full anomaly detection, and approximately 1.21 MB. For minimum footprint, the distilled student C-i at 0.894 MB offers 99.1% of the teacher’s Macro F1 with retained anomaly detection.

10. Discussion

10.1 Dual-Path Architecture Advantages

The HybridVAE demonstrates that unifying anomaly detection and classification in a single architecture is not only feasible but produces models that are operationally superior to either capability in isolation. The VAE reconstruction path and the classification encoder share no weights, allowing each to specialise without interference. The concatenation of latent features (z) with discriminative features (cls_feat) gives the classifier access to both generative and discriminative representations.

10.2 The Security Case for Anomaly Detection

The zero-day experiment provides compelling evidence that pure supervised classifiers, regardless of their accuracy on known classes, are unsuitable as standalone security tools. XGBoost's 99.98% confidence on misclassified Heartbleed samples represents a worst-case scenario: a system that not only fails to detect an attack but actively assures the operator that everything is normal. The HybridVAE's reconstruction error pathway provides a safety net that operates independently of the classification head, ensuring that novel threats trigger an alert even when the classifier has no trained category to assign.

10.3 Compression Effectiveness

The compression results demonstrate that the HybridVAE architecture contains significant redundancy that can be eliminated without sacrificing core capabilities. Knowledge distillation achieves a 3.38x parameter reduction with only 0.9% F1 degradation, while IMP at 60% sparsity actually improves Macro F1 by 2.9 percentage points. The IMP result aligns with the Lottery Ticket Hypothesis: the dense network contains a sparse sub-network that outperforms the original when found through iterative pruning and retraining.^[5]

10.4 Limitations and Future Work

Several limitations merit acknowledgement. First, the zero-day experiment uses only four test samples (Heartbleed), providing a qualitative demonstration rather than statistically robust quantification. Future work should extend this to multiple withheld classes and larger sample sizes. Second, the architecture does not incorporate temporal dependencies across network flows, which could improve detection of multi-step attack patterns. Third, actual edge deployment benchmarks (inference latency, memory footprint on ARM processors, power consumption) have not been conducted. Finally, adversarial robustness testing against evasion attacks has not been performed and represents an important direction for security-critical deployment.

11. Conclusion

This report presented HybridVAE, a semi-supervised network intrusion detection system that unifies supervised multi-class classification and unsupervised anomaly detection within a single neural network architecture. Evaluated on the CICIDS2017 benchmark, the model achieves a Macro F1-score of 0.8702 and anomaly detection AUROC of 0.8740 in a compact 2.92 MB package.

The architecture’s dual-path design proves its value most clearly in the zero-day detection experiment: while a state-of-the-art XGBoost classifier confidently misclassifies unseen Heartbleed attacks as benign traffic, the HybridVAE successfully flags all unseen samples as anomalous through its reconstruction error pathway. This capability is absent from any purely supervised approach and is critical for real-world security deployment.

Model compression through knowledge distillation and Iterative Magnitude Pruning demonstrates that the architecture can be substantially reduced for edge deployment. The distilled student retains 99.1% of the teacher’s performance at 3.38x fewer parameters, while IMP at 60% sparsity surpasses the unpruned baseline, achieving the best Macro F1 (0.899) of any HybridVAE variant. These compressed models, at under 1.3 MB, are suitable for deployment on IoT gateways, embedded controllers, and other resource-constrained edge devices where cloud connectivity is unavailable.

12. References

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Appendix A: HybridVAE Layer-by-Layer Architecture

Table A1: Detailed Architecture Specification

Component	Layer Specification	Output Dim
VAE Encoder L1	Linear(77, 512) → BN → GELU → DO(0.3)	512
VAE Encoder L2	Linear(512, 256) → BN → GELU → DO(0.3)	256
VAE Encoder L3	Linear(256, 128) → BN → GELU → DO(0.3)	128
Residual Block	Linear(128,128)→BN→GELU→DO→Linear(128,128)→BN + skip	128
μ / $\log \sigma^2$	Linear(128, 64) each	64 each
Latent z	Reparameterisation: $z = \mu + \sigma \cdot \epsilon$	64
Decoder L1	Linear(64, 128) → BN → GELU → DO(0.3)	128
Decoder L2	Linear(128, 256) → BN → GELU → DO(0.3)	256
Decoder L3	Linear(256, 512) → BN → GELU → DO(0.3)	512
Reconstruction	Linear(512, 77)	77
CLS Encoder	3× [Linear → BN → GELU → DO], dims 256→256→128	128
Classifier Input	Concatenate [z , cls_feat]	192
Classifier Head	Linear(192, 9)	9

Appendix B: Student Architecture (Knowledge Distillation)

Table B1: Teacher vs Student Dimensions

Component	Teacher	Student
Encoder layers	[512, 256, 128]	[256, 128, 64]
Latent dimension	64	32
Decoder layers	[128, 256, 512]	[64, 128, 256]
CLS encoder output	128	64
Classifier input	192 (64+128)	96 (32+64)
Total parameters	749,270	221,462
Model size	2.92 MB	0.894 MB

Appendix C: IMP Pruning Schedule

Table C1: Iterative Magnitude Pruning Configuration

Round	Cumulative Sparsity	Retrain Epochs	Learning Rate
1	20%	20	3e-4
2	35%	20	3e-4

Round	Cumulative Sparsity	Retrain Epochs	Learning Rate
3	50%	20	3e-4
4 (best)	60%	20	3e-4
5	70%	20	3e-4

Best model selected at Round 4 (60% sparsity) with Macro F1 = 0.8992 and AUROC = 0.8710.